R0 PROTOTYPE — NOT HARDWARE VERIFIED

Waveshare Core2350B2 carrier; initial scope: 32 KiB / no MBC

U1
SN74LVC245APWR
+3V3 1
GND 10
GND 19
+3V3 20
A0 2
A0_3V3 18
A1 3
A1_3V3 17
A2 4
A2_3V3 16
5 A3
15 A3_3V3
6 A4
14 A4_3V3
7 A5
13 A5_3V3
8 A6
12 A6_3V3
9 A7
11 A7_3V3

U2
SN74LVC245APWR
+3V3 1
GND 10
GND 19
+3V3 20
A8 2
A8_3V3 18
A9 3
A9_3V3 17
A10 4
A10_3V3 16
5 A11
15 A11_3V3
6 A12
14 A12_3V3
7 A13
13 A13_3V3
8 A14
12 A14_3V3
9 A15
11 A15_3V3

U3
SN74LVC245APWR
+3V3 1
GND 10
GND 19
+3V3 20
RD_L 2
RD_L_3V3 18
WR_L 3
WR_L_3V3 17
RESET_L 4
RESET_L_3V3 16
5 CLK
15 CLK_3V3
6 GND
14
7 GND
13
8 GND
12
9 GND
11

M1
Core2350B2 (31692)
GND GND1
GND GND2
GND GND3
GND GND4
A2_3V3 GP2
A4_3V3 GP4
A6_3V3 GP6
A8_3V3 GP8
A10_3V3 P10
A12_3V3 P12
A14_3V3 P14
GND GND7
A1_3V3 GP1
A3_3V3 GP3
A5_3V3 GP5
A7_3V3 GP7
A9_3V3 GP9
A11_3V3 P11
A13_3V3 P13
A15_3V3 P15
D1_3V3 GP17
D3_3V3 GP19
D5_3V3 GP21
D7_3V3 GP23
WR_L_3V3 P25
CLK_3V3 P27
RESET_HOLD
GP30
D0_3V3 GP16
D2_3V3 GP18
D4_3V3 GP20
D6_3V3 GP22
GP24 RD_L_3V3
GP26 RESET_L_3V3
GP28 VALID
GP32
GP34
GP36
GP38
GP40
GP42
GP44
GP45
GP31
GP33
GP35
GP37
GP39
GP41
GP43
GP0 A0_3V3
VBUS MOD_VBUS
3V3_EN
RUN
USB_DP
USB_DM
BOOTSEL
GP47
3V3 +3V3
GND6 GND
ADC_VREF
SWCLK
SWD
GP46

U4
SN74LVC8T245PWR
+3V3 1
+3V3 2
GND 11
GND 12
GND 13
OE_L 22
+5V_GB 23
+5V_GB 24
D0_3V3 3
D0 21
D1_3V3 4
D1 20
5 D2_3V3
19 D2
6 D3_3V3
18 D3
7 D4_3V3
17 D4
8 D5_3V3
16 D5
9 D6_3V3
15 D6
10 D7_3V3
14 D7

U5
SN74LVC1G10DBVR
VALID 1
GND 2
WR_L_3V3 3
4 INHIBIT
5 +3V3
RESET_L_3V3 6

U6
SN74LVC1G332DBVR
INHIBIT 1
GND 2
RD_L_3V3 3
4 OE_L
5 +3V3
6 A15_3V3

Read enable: /OE = /RD OR A15 OR NOT(VALID AND /WR AND /RESET)

VALID is cleared for every read, then asserted only after correct data is loaded.

PWR_FLAG MOD_VBUS

Q1
2N7002
RESET_HOLD 1
GND 2
3 RESET_L

D1
SS14
MOD_VBUS 1
2 +5V_GB

R1
4.7k
VALID 1
2 GND

R2
10k
+3V3 1
2 RESET_HOLD

R3
10k
+3V3 1
2 OE_L

J1
GAME BOY 32 CONTACTS
+5V_GB 1
CLK 2
WR_L 3
RD_L 4
5
A0 6
A1 7
A2 8
A3 9
A4 10
A5 11
A6 12
A7 13
A8 14
A9 15
A10 16
+5V_GB
CLK
WR_L
RD_L
CS_L
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
D0
D1
D2
D3
D4
D5
D6
D7
RESET_L
VIN
GND
17 A11
18 A12
19 A13
20 A14
21 A15
22 D0_GB
23 D1_GB
24 D2_GB
25 D3_GB
26 D4_GB
27 D5_GB
28 D6_GB
29 D7_GB
30 RESET_L
31
32 GND

C1
100nF
+3V3 1
2 GND

C2
100nF
+3V3 1
2 GND

C3
100nF
+3V3 1
2 GND

C4
100nF
+3V3 1
2 GND

C5
100nF
+5V_GB 1
2 GND

C6
100nF
+3V3 1
2 GND

C7
100nF
+3V3 1
2 GND

C8
10uF
MOD_VBUS 1
2 GND

R4
33
D0 1
2 D0_GB

R5
33
D1 1
2 D1_GB

R6
33
D2 1
2 D2_GB

R7
33
D3 1
2 D3_GB

R8
33
D4 1
2 D4_GB

R9
33
D5 1
2 D5_GB

R10
33
D6 1
2 D6_GB

R11
33
D7 1
2 D7_GB

Module: hand-assembly. J1: PCB fingers. All other parts: PCBA.

Electrical / mechanical validation required

Sheet: /
File: romcart.kicad_sch

Title: RP2350B ROM cartridge — Core2350B2 carrier

Size: A2	Date: 2026-10-12	Rev: R0 prototype
KiCad E.D.A. 10.0.6		Id: 1/1